

Real-time Seamless Multi-Projector Displays on Deformable Surfaces

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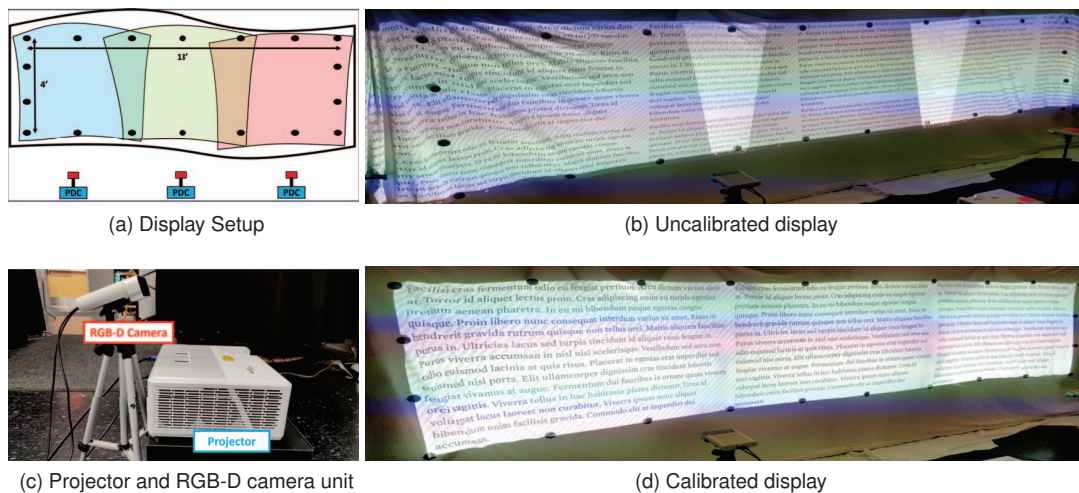


Figure 1: (a) Illustration of our setup, comprising 3 PDCs with overlapping projection covering a 18' × 4' deformable surface. (b) The uncalibrated display without any warp and blend. (c) A projector and RGB-D camera. (d) The final seamless display.

ABSTRACT

Prior work on multi-projector displays have focused primarily on rigid objects. However, works on projection based displays on deformable dynamic objects have focused only on small scale single projector systems. Tracking a deformable dynamic surface and updating projections precisely in real time to project on it is a significantly challenging task, even for single projector systems.

For my dissertation, I am working on developing the first end-to-end solution for achieving a real-time, seamless display on deformable surfaces using multiple projectors without requiring any prior knowledge of the surface shape or device calibration parameters. Challenges in achieving such a system include accurate calibration of the devices despite a continuously deforming surface, real-time tracking of the changing surface shape using multiple cameras, and real-time warp and blend of the projected imagery to create the seamless display on it.

This work has tremendous applications on mobile and expeditionary systems where environmental factors (e.g. wind, vibrations, suction) cannot be avoided. One can create large displays on tent walls in remote, austere military or emergency operations in minutes to support large scale command and control, mission rehearsal, large scale visualization or training operations. It can be used to create displays on mobile and inflatable objects for trade shows/events and touring entertainment applications.

Index Terms: Spatially Augmented Reality—Computer Vision—

1 INTRODUCTION

Multi-projector displays are an easy way to create seamless displays at scale without undertaking massive engineering feats of tiling LCD/LED panels together. Though multi-projector displays on

rigid objects (static or dynamic) have been explored extensively, expeditionary or mobile systems often demand such displays on surfaces that continuously move and deform. Examples include expeditionary tents (for defense and emergency management operations) in austere locations for large data visualizations in command and control, decision making, mission rehearsal and even training applications. However, such fabric based surfaces suffer continuous movement and deformations from environmental factors like wind and vibrations. Similar situations are faced in the commercial domain where mobile and inflatable displays need to be set up on tradeshow floors in the presence of HVAC systems blowing air or vibrations nearby from other demonstrations, causing the display surface to undergo continuous deformations. Mobile educational systems (e.g. mobile planetariums) that travel to serve geographically under-served populations can also benefit from displays on moving deformable surfaces.

Problem: The easiest way to set up such displays is to *tile* multiple projectors on the walls of these tents to create a large high-resolution display. My dissertation aims to create a legible and seamless display on such dynamic deformable surfaces that can scale to multiple projectors. We aspire to avoid long drawn calibrations with fiducials and additional 3D or 2D objects (e.g. checkerboard or patterned 3D cube) and therefore assume that the projectors and cameras are not calibrated a priori.

Challenges and Related Work: All *multi-projector* systems explored so far focus on rigid objects, either static [8, 11] or dynamic [7, 10]. Any rigid object offers constant surface normals and no local deformation of the surface. Therefore, if the 3D geometry of the object can be determined and the projectors and cameras calibrated a priori, dynamic projection mapping can be achieved [9]. Some prior work exist that use coaxial projector-camera systems (that does not need 3D calibration) or invisible inks to achieve single projector displays on deformable dynamic surfaces [5, 6] but none address multiple projector systems.

The primary challenge in a dynamic deformable system is the

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real-time estimation of the changing object shape. In order to enable this, we use RGB-D cameras (e.g. Kinect) instead of RGB cameras. These provide registered, low resolution depth in addition to a high resolution RGB image. However, to enable a scalable solution, we use multiple RGB-D cameras since the entire projection surface may not be visible to a single camera. Multiple projectors and RGB-D cameras are connected to a PC server to create the multi-projector display (see Figure-1).

Our goal is to create a seamless display in which the projected image "sticks" to the rigid object as it moves. The precision required in any tiled and/or stacked multi-projector system, even when rigid and static, is quite exacting. If the physical set up and its estimation (via calibration) do not match, the display suffers from visible misregistrations. Therefore, multi-projection registration on dynamic deformable object is extremely challenging, especially when considering high resolution content like text.

Though blending in the overlapping regions can be leveraged to help the situation, effective blending solutions assume static surface normals and therefore rigid objects. Surface normals change continuously on dynamic deformable surfaces (e.g. a moving fabric) leading to large local deformations. Devices are also not synchronized to enable consistent tracking from multiple RGB-D cameras in the overlap region. Therefore, this renders most of the current techniques (e.g. calibration, blending) to be not applicable.

2 PROPOSED WORK

My dissertation aims to develop scalable algorithms to create a seamless legible display from multiple asynchronous projectors on a large deformable, dynamic surface, without any prior knowledge of the surface shape, system calibration or device synchronization. Following are the different aspects of the system being addressed by my dissertation.

2.1 System Calibration

Accurate system calibration is the first step in achieving any seamless multi-projector display. System calibration refers to determining the device parameters (i.e. intrinsics, distortion and pose) of every projector and RGB-D camera comprising the display. High calibration accuracy is crucial to ensure registration in the regions of overlap of projectors, especially since the display surface is dynamic and deformable. Therefore, prior works on calibration that assume a static and rigid display surface become inadequate in this situation [4, 11].

Current Progress: Recently, we proposed first calibration method that exploits the surface movement to accurately calibrate a multi-projector multi-RGB-D camera system [1] without using any specific calibration pattern (e.g. checkerboard). We use multiple RGB-D cameras (need not be synchronized) to capture multiple frames of a grid of ArUCo patterns projected by a projector on the deformable surface as it is moving, one projector at a time. Next, we establish 2D-pixel to 3D-point correspondences between the projector and camera pixels across those frames. However, unlike traditional calibration methods that assume static object, we use ray-plane based constraints imposed by the pinhole camera model to calibrate the RGB-D cameras. This information is then utilized to calibrate the projectors to achieve full system calibration.

2.2 Real-time Deformable Surface Tracking

The next challenge is to reconstruct the entire deformable surface shape from the multiple calibrated cameras in real-time. Since each RGB-D camera is observing only part of the deforming surface, the depth data from each camera must be combined in real-time and aligned in a global coordinate system. Accuracy of the reconstructed shape is important, as errors will manifest as visible mis-registrations in the final projected display even with accurate calibration. Depth data from these cameras is quite noisy making this process particularly challenging.

Current Progress: Using the camera calibration parameters, I combine the surface shape from multiple cameras in real-time and

merge the overlapping regions. I sample 3D points across the entire surface to get an accurate estimate of the shape. These 3D points and the projector calibration parameters are used to compute warp maps for each projector in the system [2].

2.3 Real-time Multi-Projector Blending

The warp maps will enable a registered display, but cannot compensate for the higher brightness in the projector overlapping regions. For this, the projector intensities must be blended smoothly across the overlapping regions. Unlike prior work that assumes static objects for blending, for a constantly changing surface, the blend masks must adapt in real-time to the changing surface shape [2].

Current Progress: I have designed a novel blending method that computes the blending weights at every 3D coordinate of the display based on the distance to the nearest view frustum plane. If a 3D point is close to a plane, then its blend weight in the corresponding projector is reduced. Conversely, if it is far, then its weight is increased. Our method is very efficient and currently takes 22ms to render a complete display frame (less than the 33ms necessary for real time display) [2].

3 FUTURE PROBLEMS TO BE ADDRESSED

Our first system that achieves real-time seamless multi-projector displays on deformable surfaces [2]. However, some additional challenges remain. Consumer-grade depth cameras provide noisy depth maps that impact both calibration accuracy and real-time shape recovery, resulting in visible mis-registrations and jitter. I plan to use smoothing methods such as the Kalman filter to help reduce the noise and improve display quality. Further, misregistrations can be the consequence of the lag introduced by the shape reconstruction, warp map and blend mask generation steps. This is because by the time the updated projection is rendered, the surface shape has changed slightly. Real-time motion prediction methods could be used to alleviate this problem. Finally, we could explore leveraging real-time content-dependent blending methods [3] to further hide misregistrations artifacts.

REFERENCES

- [1] M. T. Ibrahim, M. Gopi, and A. Majumder. Auto-calibrating dynamic projection mapping system with jitter and occlusion management. In *2023 IEEE ISMAR*.
- [2] M. T. Ibrahim, M. Gopi, and A. Majumder. Real-time seamless multi-projector displays on deformable surfaces. In *Currently under Review*.
- [3] P. Kurth, M. Leuschner, M. Stamminger, and F. Bauer. Content-aware brightness solving and error mitigation in large-scale multi-projection mapping. *IEEE TVCG*, 2022.
- [4] D. Moreno and G. Taubin. Simple, accurate, and robust projector-camera calibration. In *2012 3D-MPVT*.
- [5] G. Narita, Y. Watanabe, and M. Ishikawa. Dynamic projection mapping onto a deformable object with occlusion based on -speed tracking of dot marker array. In *ACM VRST*, 2015.
- [6] G. Narita, Y. Watanabe, and M. Ishikawa. Dynamic projection mapping onto deforming non-rigid surface using deformable dot cluster marker. *IEEE TVCG*, 2016.
- [7] T. Nomoto, W. Li, H. Peng, and Y. Watanabe. Dynamic multi-projection mapping based on parallel intensity control. *IEEE TVCG*, 2022.
- [8] B. Sajadi, M. A. Tehrani, M. Rahimzadeh, and A. Majumder. High-resolution lighting of 3d reliefs using a network of projectors and cameras. In *2015 3DTV-Conference*.
- [9] C. Siegl, M. Colaïanni, M. Stamminger, and F. Bauer. Stray-light compensation in dynamic projection mapping. In *SIGGRAPH ASIA*, 2016.
- [10] C. Siegl, M. Colaïanni, L. Thies, J. Thies, M. Zollhöfer, S. Izadi, M. Stamminger, and F. Bauer. Real-time pixel luminance optimization for dynamic multi-projection mapping. In *ACM TOG*, 2015.
- [11] M. A. Tehrani, M. Gopi, and A. Majumder. Automated geometric registration for multi-projector displays on arbitrary 3d shapes using uncalibrated devices. *IEEE TVCG*, 2019.